3.8.5 Word problems that involve linear equations

The challenging part of solving word problems is translating the statements into equations. Collect as many such problems and attempt to solve them.

Example 3.33

The sum of two numbers is 36 and one number exceeds another by 8. Find the numbers.

Solution:

Let the smaller number be x and the greater number be $x+8$

Given: the sum of two numbers = 36

$$\begin{aligned} x + (x+8) &= 36 \\ 2x + 8 &= 36 \\ 2x &= 36 - 8 \\ 2x &= 28 \\ x &= \frac{28}{2} = 14 \end{aligned}$$

Hence,

(i) The smaller number, $x = 14$

(ii) The greater number, $x + 8 = 14 + 8 = 22$

Example 3.34

A bus is carrying 56 passengers with some people having ₹8 tickets and the remaining having ₹10 tickets. If the total money received from these passengers is ₹500, find the number of passengers with each type of tickets.

Solution:

Let the number of passengers having ₹8 tickets be y. Then, the number of passengers with ₹10 tickets is (56–y).

Total money received from the passengers = ₹500

That is,
$$\begin{aligned} y \times 8 + (56 - y) \times 10 &= 500 \\ 8y + 560 - 10y &= 500 \\ 8y - 10y &= 500 - 560 \\ -2y &= -60 \\ y &= \frac{60}{2} \\ y &= 30 \end{aligned}$$

Hence, the number of passengers having,

(i) ₹8 tickets = 30

(ii) ₹10 tickets = 56–30 = 26

Example 3.35

The length of a rectangular field exceeds its breadth by 9 metres. If the perimeter of the field is 154m, find the length and breadth of the field.

Solution:

Let the breadth of the field be 'x' metres; then its length (x+9) metres.

Perimeter of the P = 2(length + breadth) = $2(x + 9 + x) = 2(2x + 9)$

Given that, $2(2x + 9) = 154$.

$$\begin{aligned} 4x + 18 &= 154 \\ 4x &= 154 - 18 \\ 4x &= 136 \\ x &= 34 \end{aligned}$$

Hence,

(i) Thus, breadth of the rectangular field = 34m

(ii) length of the rectangular field = $x+9 = 34+9 = 43m$

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